Alture AI: A Hybrid Multi-Modal NLP Architecture for Explainable ATS Resume--Job Compatibility and Reciprocal Recommendation

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Abstract*—Applicant Tracking Systems (ATS) serve as the primary automated gatekeepers in modern corporate recruitment pipelines. However, conventional ATS algorithms predominantly rely on syntactic keyword heuristics or isolated dense embeddings, suffering from vocabulary mismatch, inability to capture non-linear skill interactions, and an absence of actionable explainability for job applicants. This paper presents Alture AI, a hybrid multi-modal matching framework that unifies four complementary signal streams: (1) dense semantic representations generated via Sentence-BERT (all-MiniLM-L6-v2), (2) pairwise cross-encoder token-level attention interaction, (3) an alias-normalized 500+ technical skill ontology computing explicit Jaccard overlap, coverage recall, and gap penalty metrics, and (4) document structural complexity features including length ratios and Type-Token Ratios (TTR). These heterogeneous features are calibrated through a multi-task classification head and fused via tuned gradient-boosted decision ensembles (XGBoost, LightGBM, CatBoost) and a Stacking Meta-Learner. Evaluated on the Resume-ATS Score Dataset v1 (6,374 pairs), Alture AI achieves an R^2 of 0.3164, a top-tier shortlist precision of 70.86%, and a ranking quality of 96.36% nDCG@10, significantly outperforming lexical and single-signal baselines. The system is deployed as an asynchronous FastAPI microservice on Hugging Face ZeroGPU with an interactive Vercel React client, providing transparent, explainable skill-gap diagnostics and generative career coaching.

Index Terms—**Applicant Tracking Systems, Natural Language Processing, Sentence-BERT, Explainable AI, Job Recommendation, Gradient Boosting, XGBoost.**

I. INTRODUCTION

Algorithmic screening of curriculum vitae (CVs) has become an indispensable foundation of modern talent acquisition. With international job openings routinely attracting hundreds of applicants within hours of posting, manual resume auditing has become intractable for enterprise recruiters. Automated Applicant Tracking Systems (ATS) filter, score, and rank candidate profiles against job descriptions before human recruiters review them.

Despite widespread industrial adoption, traditional ATS platforms suffer from fundamental architectural flaws. First, legacy keyword indexing systems fail to capture semantic equivalence, penalizing qualified applicants who describe identical proficiencies with alternate nomenclature (e.g., PyTorch vs. deep neural network development). Second, pure bi-encoder dense embedding models produce coarse global similarity scores that gloss over non-negotiable prerequisite skills, leading to hallucinations of qualification.

To overcome these limitations, we introduce **Alture AI**, an end-to-end explainable matching engine that synthesizes dense contextual embeddings, cross-encoder token attention, a 500+ technical skill ontology, and structural document metrics within a gradient-boosted ensemble stack. Our system balances holistic semantic alignment with strict domain-level verification, delivering actionable feedback to candidates alongside rigorous candidate ranking for employers.

II. RELATED WORK

Research in automated resume-job matching spans three primary paradigms: lexical matching, dense representation learning, and reciprocal graph matching. We review five benchmark contributions below.

1. Panchasara et al. (2023): Evaluated transformer-based embeddings for recruitment recommendation, demonstrating that BERT models significantly outperform legacy TF-IDF representations. However, their architecture operated solely on global semantic similarity, omitting explicit verification of compulsory technical proficiencies.

2. Zhu et al. (2021): Formulated recruitment as a bipartite graph optimization problem, introducing dual-perspective reciprocal constraints. While effective for platform equilibrium, their model treated text as monolithic latent vectors without interpretable feature attribution.

3. Mao et al. (2023): Proposed tensor decomposition combined with multi-head attention to capture higher-order applicant-job interactions. Their approach achieved competitive ranking precision but suffered from quadratic inference latency unsuitable for real-time web deployment.

4. Guan et al. (2024): Introduced JobFormer, integrating skill entity nodes into transformer self-attention layers. Their findings confirm that explicit skill awareness prevents semantic drift in long job descriptions, inspiring our hybrid ontology approach.

5. Han et al. (2024): Addressed temporal drift in candidate preference modeling using behavioral-semantic fusion learning. Their study highlighted the necessity of structural document features in regularizing semantic similarity scores.

III. DATASET & PREPROCESSING

We validate our architecture on the **Resume-ATS Score Dataset v1 (English)** curated by 0xnbk, comprising 6,374 validated resume-job description pairs with continuous compatibility scores scaled in [0, 100].

Data Cleaning: Text pipelines apply unicode normalization, regex whitespace compaction, email/URL de-identification, and HTML artifact elimination. Token statistics show average resume lengths of 342 words (range 48-1,142) and job description lengths of 218 words (range 35-780).

Splitting Protocol: The corpus is partitioned into an 80% training set (5,099 pairs) and a held-out 20% test set (1,275 pairs) using fixed seed stratification across target score quartiles.

IV. METHODOLOGY

Alture AI computes a composite 784-dimensional multi-modal feature vector combining four heterogeneous signal streams:

1. Dense Semantic Embeddings: Resumes and job descriptions are mapped into 384-dimensional dense vectors using all-MiniLM-L6-v2 Sentence-BERT. Pairwise similarity is computed via cosine similarity and Manhattan distance.

2. Cross-Encoder Token Attention: Joint resume-job token interaction is calibrated via cross-encoder classification heads, capturing contextual token interplay across qualification boundaries.

3. 500+ Technical Skill Ontology: A curated ontology spanning Languages, Frameworks, Cloud, Databases, and MLOps extracts matched skills S_R cap S_J and missing prerequisites S_J minus S_R, computing Jaccard index and coverage recall.

4. Document Structural Complexity: Measures Type-Token Ratio (TTR), word count ratios, and bullet density to model document completeness.

5. Gradient-Boosted Ensemble Stack: Level-0 base estimators (Ridge, Random Forest, LightGBM, CatBoost, Tuned XGBoost) feed out-of-fold predictions into a meta-regressor, yielding calibrated final score predictions.

V. EXPERIMENTAL RESULTS

We compare Alture AI against standard baselines across regression accuracy (RMSE, MAE, R^2), ranking quality (nDCG@10), and top-tier shortlist precision (Precision@GoodFit):

Model Architecture	RMSE v	MAE v	R^2 ^	nDCG@10 ^	Precision ^
TF-IDF + Ridge	18.42	14.85	0.082	82.14%	46.20%
TF-IDF + Random Forest	16.94	13.20	0.198	87.50%	55.40%
SBERT + Ridge	16.12	12.48	0.245	91.20%	62.80%
Hybrid + LightGBM	15.54	11.85	0.298	94.80%	68.10%
Hybrid + Tuned XGBoost	15.31	11.62	0.3164	96.36%	70.86%
Stacking Ensemble (Meta)	15.24	11.55	0.3220	96.80%	71.50%

Table I: Comprehensive Model Evaluation Benchmark on Resume-ATS Score Test Split (1,275 pairs).

VI. DISCUSSION & DIAGNOSTICS

Signal Synergies: Dense contextual embeddings provide macro semantic alignment, while the 500-entity ontology provides micro-level hard skill verification, eliminating false positive hallucinations.

Failure Modes: Resumes with fewer than 80 words exhibit increased residual variance due to sparse token density. Highly novel or proprietary tooling absent from the ontology falls back onto dense semantic matching.

Ethical Safeguards: Demographic metadata (gender, age, location, university prestige) is stripped prior to feature engineering to guarantee meritocratic ranking.

VII. CONCLUSION & DEPLOYMENT

Alture AI successfully unifies multi-modal semantic NLP with explicit ontology-driven verification. The complete system is deployed as an asynchronous FastAPI backend on Hugging Face ZeroGPU with an interactive React frontend on Vercel, providing real-time ATS screening, explainable skill diagnostics, and automated career coaching.

Future work will extend our framework to multilingual job matching and graph neural networks for reciprocal candidate-employer interaction modeling.

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